NTIRE 2026 Image Super-Resolution (×4) Challenge Factsheet –Image Super-Resolution Reconstruction Using ProCata-

Shuling Zheng
Jinan University
Guangzhou, China
zhengshuling@stu2025.jnu.edu.cn

Zhiheng Fu
Jinan University
Guangzhou, China
fuzhiheng@stu2025.jnu.edu.cn

Feng Zhang
Jinan University
Guangzhou, China
zhangfeng@stu2025.jnu.edu.cn

Zhanglu Chen
Jinan University
Guangzhou, China
czhanglu@stu2024.jnu.edu.cn

Boyang Yao
Jinan University
Guangzhou, China
yaoboy@stu2024.jnu.edu.cn

1. Team details

- Team name: JNU620
- Team leader name: Shuling Zheng
- Team leader address, phone number, and email: Jinan University, Guangzhou, 13822956608, zhengshuling@stu2025.jnu.edu.cn
- Rest of the team members: Zhiheng Fu, Feng Zhang, Zhanglu Chen, Boyang Yao
- Affiliation: Jinan University
- User names and entries on the NTIRE 2026 Codabench competitions: shuling
- Best scoring entries of the team during the testing phase: PSNR=30.47 SSIM=0.8515
- Link to the codes/executables of the solution(s): https://github.com/zslstar/NTIRE2026_ImageSR_x4.git

2. Method details

2.1. Network

Inspired by CATANet[2] and DPA[1], we take CATANet as our baseline model and design a improved network named ProCata. The overall framework of ProCata is illustrated in Figure 1, which mainly consists of Aggregation Blocks, Region-Aware Attention (RAA) , and Dual Pool Attention (DPA). We design an Aggregation Block (AB) based on TAB in CATANet[2]. Inside the AB, three depthwise separable convolutions are employed to aggregate multi-scale information, and residual connections are adopted to preserve and combine shallow-level features. Furthermore, we use ECA[3] to design a Region-Aware Attention (RAA) mechanism improved from the LRSA in the original CATANet[2] . Our region-aware attention is capable of focusing on both local and global details, leading to clearer and more faithful restoration results.

2.2. Training strategy

We train our model on DIV2K training dataset, which consists of paired high-resolution (HR) and low-resolution (LR) images. All images are cropped to 256×256 pixels, and data augmentation is performed via random horizontal flipping and rotation. We use a batch size of 16 and repeat the dataset 100 times. The model is optimized with the Adam optimizer at a learning rate of 1×10 for a total of 250,000 iterations, employing the L1 loss as the training objective.

2.3. Experimental results

we evaluate our model on DIV2K test dataset: PSNR=30.47, SSIM=0.8515

References

[1] Qian Jiang, Qianqian Wang, Xin Jin, Michal Wozniak, Shaowen Yao, and Wei Zhou. Mfmamba: A multi-function network for panchromatic image resolution restoration based on state-space model. 2025. 1

[2] Xin Liu, Jie Liu, Jie Tang, and Gangshan Wu. Catanet: Efficient content-aware token aggregation for lightweight image super-resolution. 2025. 1

[3] Qilong Wang, Banggu Wu, Pengfei Zhu, Peihua Li, and Qinghua Hu. Eca-net: Efficient channel attention for deep convolutional neural networks. *IEEE*, 2020. 1

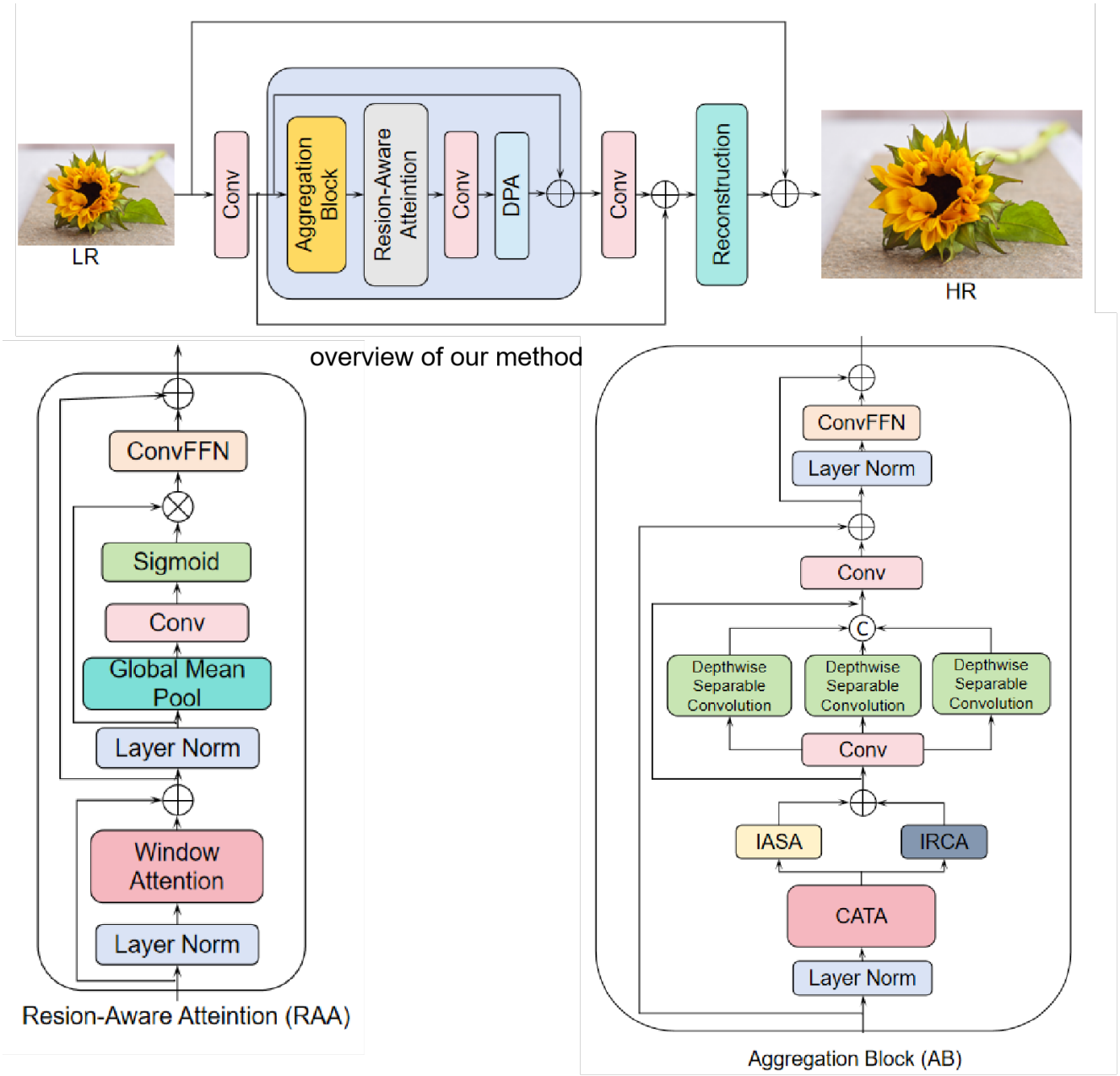

Figure 1. The Network architecture of ProCata